

1. Purpose

This spike validates Paged.js pagination capabilities for the DocMaker architecture decision. The purpose paragraph is intentionally long enough to interact with pagination behaviour in a realistic way. It repeats: controlled laboratory documentation requires deterministic page-oriented rendering with headers, footers, page numbering, and repeated table headers across page boundaries.

1.1 Background

Background paragraph one. Quality manuals, standard operating procedures and work instructions all rely on stable pagination. This paragraph pads the page so the following table begins near the page boundary, which is required to observe multi-page table behaviour.

Background paragraph two with additional filler content so that the table below starts close to the bottom of page one. Repeat filler: accreditation documentation, calibration records, environmental monitoring, corrective action registers, internal audit schedules and management review minutes.

Table 1 — Sample Acceptance Criteria

Parameter	Method	Limit	Result	Remark
pH	IS 3025 Pt. 40	6.5 – 8.5	7.2	Pass
Turbidity	IS 3025 Pt. 21	< 1 NTU	0.4	Pass
TDS	IS 3025 Pt. 26	< 500 mg/L	312	Pass
Chloride	IS 3025 Pt. 23	< 250 mg/L	88	Pass
Hardness	IS 3025 Pt. 19	< 200 mg/L	142	Pass
Iron	IS 3025 Pt. 59	< 0.3 mg/L	0.11	Pass
Fluoride	IS 3025 Pt. 60	< 1.0 mg/L	0.42	Pass
Sulphate	IS 3025 Pt. 30	< 200 mg/L	96	Pass
Nitrate	IS 3025 Pt. 34	< 45 mg/L	7.9	Pass
Colour	IS 3025 Pt. 2	< 15 TCU	4	Pass
Odour	Organoleptic	Acceptable	Nil	Pass
EC	IS 3025 Pt. 302	< 1500 µS/cm	640	Pass
Calcium	IS 3025 Pt. 45	< 100 mg/L	58	Pass
Magnesium	IS 3025 Pt. 45	< 100 mg/L	29	Pass
Sodium	IS 3025 Pt. 47	< 200 mg/L	64	Pass
Potassium	Flame photometry	< 50 mg/L	12	Pass
Zinc	AAS	< 5.0 mg/L	0.08	Pass
Copper	AAS	< 1.5 mg/L	0.03	Pass
Manganese	AAS	< 0.1 mg/L	0.02	Pass
Lead	AAS	< 0.01 mg/L	< 0.01	Pass

Parameter	Method	Limit	Result	Remark
Arsenic	Hydride generation	< 0.01 mg/L	< 0.01	Pass
Cadmium	AAS	< 0.003 mg/L	< 0.003	Pass
Nickel	AAS	< 0.02 mg/L	0.005	Pass
Chromium	AAS	< 0.05 mg/L	0.01	Pass
Cyanide	IS 3025 Pt. 42	< 0.05 mg/L	< 0.02	Pass
Phenol	4-AAP	< 0.001 mg/L	< 0.001	Pass
MBAS	IS 3025 Pt. 15	< 0.2 mg/L	0.05	Pass
Residual Cl	DPD	< 0.2 mg/L	0.1	Pass
Coliform	MPN	Absent	Absent	Pass
E. coli	MPN	Absent	Absent	Pass
Total plate count	IS 3025 Pt. 90	< 100 CFU/mL	< 1	Pass
Halogens	Titration	< 200 mg/L	121	Pass
Alkalinity	Titration	< 200 mg/L	150	Pass
Free CO ₂	Titration	Report	8	Info
Oil & grease	Hexane extract	< 10 mg/L	< 1	Pass
SS	Gravimetric	< 100 mg/L	22	Pass
BOD	IS 3025 Pt. 58	< 30 mg/L	4	Pass
COD	Reflux titration	< 100 mg/L	18	Pass
Ammonia	Phenate	< 0.5 mg/L	0.12	Pass
Hardness CaCO ₃	Titration	< 200 mg/L	142	Pass
Moisture	Karl Fischer	Report	0.04%	Info
Conductivity	IS 3025 Pt. 302	< 1500	640	Pass
Particle count	ISO 11500	Report	Class 8	Info
Viscosity	Brookfield	Report	1.01 cP	Info
Flash point	Pensky-Martens	Report	N/A	Info
Density	pycnometer	Report	0.998	Info
Melting point	Capillary	Report	N/A	Info
Assay	Titration	99.5%	99.7%	Pass
Loss on drying	Gravimetric	< 0.5%	0.21%	Pass
Residue on ignition	Gravimetric	< 0.1%	0.03%	Pass
Heavy metals	ICP-MS	< 10 ppm	< 1	Pass

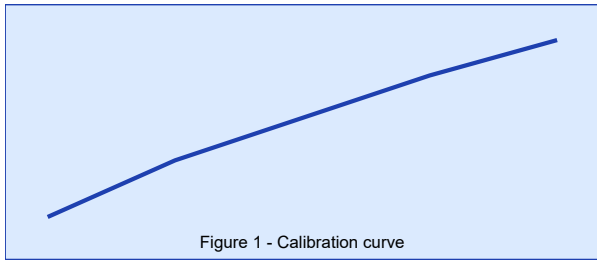


Figure 1 — Instrument calibration curve (data-URI image)

Keep-together paragraph A. These two paragraphs must never be split across a page boundary because the layout engine honours break-inside: avoid on their container.

Keep-together paragraph B. If pagination places paragraph A on one page, paragraph B must appear on the same page or the whole block must move to the next page.

1.2 Records

Records paragraph after the keep-together block, used to observe whether the avoid-break block behaves sensibly when followed by more content near the page boundary.

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2. Landscape Section

This section must render as A4 landscape with its own header and footer rules.

Equipment	Id	Range	Cal due	Std	Traceability	Location	Status
Balance	EQ-01	0–220 g	2026-11-01	Class E2	NPL	Lab 1	OK
Pipette	EQ-07	1–10 mL	2026-12-15	Gravimetric	NPL	Lab 1	OK
Oven	EQ-11	ambient–300 °C	2027-01-20	PT-100 probe	NPL	Lab 2	OK
pH meter	EQ-03	0–14 pH	2026-10-30	Buffer 4/7/9	NPL	Lab 1	OK
Conductivity	EQ-04	0–200 mS	2026-11-11	KCl std	NPL	Lab 1	OK
Incubator	EQ-09	20–60 °C	2027-02-01	PT-100 probe	NPL	Lab 3	OK

3. Back to Portrait

The final section returns to A4 portrait and reuses the running header pattern with "Page X of Y" in the footer.

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